Jiaqi (Charles) Shang

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SUMMARY

Research-driven ML scientist focused on Generative Modelling, Tabular Learning, Computational Imaging, Speech Enhancement, and Causality. Led EDA and experiments across OpenML/T4; advanced transformer-based tabular modeling and intermediate-step compositional attention; and designed a contextual RBF histogram policy for bandits. Comfortable owning the full cycle (EDA, training, evaluation) and translating findings into usable tools with cross-lab and industry partners.

EDUCATION

University of Toronto, *Toronto, Ontario, Canada* Sept. 2025 - Dec. 2026 (Expected)
Master of Science in Applied Computing

- **Coursework:** Computational Imaging, Machine Learning, Causal Inference, Computational Models of Semantic Change.

University of Waterloo, *Waterloo, Ontario, Canada* Sept. 2020 - Apr. 2025
Bachelor of Mathematics in Computer Science & Statistics, Minor in Computational Mathematics

- GPA: 90.36%, obtained Academic Excellent Standing and Term Distinction for all terms.
- **Coursework:** Artificial Intelligence, Machine Learning, Algorithms and Data Structure, Computational Inference.

WORK EXPERIENCE

Full Stack Developer (Co-op), *Aterlo Networks, Waterloo, Ontario, Canada* May. 2023 - Dec. 2023

- Engineered OOP **Python** data models and a cloud ingestion pipeline for multi-agent network telemetry, supporting 10M+ fixed-wireless/fiber devices across 20+ customer sites and reducing network load by 15%.
- Automated unit-test generation with **Python unittest and mock** (diff detection and docs) for thousands of models, accelerating regression coverage and team adoption; tool was integrated into workflows.
- Designed and shipped a responsive device information card UI (**JS/TS/Node, CSS**) surfacing port stats and connected-device counts across all supported routers/APs, validated via simulation on phone/tablet/desktop.
- Implemented semaphore-based session locking for **SSH** device tooling to prevent concurrent writes, stabilizing **APIs** under load and cutting failure modes from overlapping sessions.
- Built a **Protobuf** to **UML** generator and wired it into **Bitbucket CI**, visualizing internal **API message** relationships to speed schema reviews and clarify service contracts across **Python/JS** codebases.
- Reverse-engineered legacy devices (no manuals) using **cURL** and vendor CLIs; instrumented **Google Cloud queries** and dashboards to isolate model-specific logic defects and shorten debug cycles.

Data Engineer (Co-op), *Tesla, Shanghai, China* Mar. 2022 - Aug. 2022

- Built an end-to-end **Django** and **MySQL** portal for recruiting data (1K+ rows, 40+ fields, 20 users); added **SSO**, **RBAC**, and row-level access to eliminate overwrites and lift team throughput 50%.
- Implemented **CRUD**, advanced **SQL** filters, and **Bokeh** interactive tables/charts; automated sub-team weekly summaries with email digests, saving hours of manual reporting each week across 5 groups.
- Containerized with **Docker** and deployed on **Kubernetes**; delivered reproducible dev/prod environments and reliable releases while coordinating access/compliance with 3+ departments.
- Built **PySpark** and **Airflow ELT** joining vision and telematics for seat-belt detection analysis; disentangled true unbuckled behavior from detector failures, flagging 20% of 10M+ events as potential model misses.
- Published curated **MySQL** gold tables and SQL-ready reports powering safety analytics and offline model evaluation; accelerated root-cause triage and focused model iteration on highest-impact scenarios.
- Bootstrapped **Excel/VBA macros** (data validation, dropdowns, summary generation) to stabilize the legacy sheet pre-migration, reducing data-entry errors and transition risk.

Machine Learning Engineer (Co-op), *iFlytek, Beijing, China* Sept. 2021 - Dec. 2021

- Engineered an ensemble of seven Chinese paraphrase-generation algorithms (**Python, jieba, LTP**), delivering 95.8% precision and 99.2% recall while preserving fluency and maximizing semantic diversity.
- Built a **BERT back-translation** pipeline to augment training data and diversify candidates, strengthening model generalization across domains.
- Preprocessed 1M+ sentences with generative models via a scalable, reproducible data pipeline, enabling large-scale training and evaluation.
- Applied semantic role labeling, POS/NER, and dependency parsing (**LTP**) to enforce linguistic constraints during generation, improving readability and grammaticality of outputs.
- Established an evaluation framework tracking precision and recall, creating objective quality gates for release decisions.

- Synthesized and presented state-of-the-art research on paraphrase generation and semantic similarity, informing algorithm selection and prioritizing high-impact experiments.

RESEARCH EXPERIENCE

Undergraduate Research Assistant, *University of Waterloo, Waterloo, Ontario, Canada* Jan. 2024 - Apr. 2025
Supervisor: Pascal Poupart, University of Waterloo

- Co-designed a **transformer-based** tabular architecture ("basis transformers") to handle heterogeneous columns, text features, and missing or noisy values; lifted median R^2 by +0.338 with the lowest variance across 34 OpenML-CTR23 tasks while using 5x fewer parameters than the best baseline.
- Performed end-to-end **EDA** on **T4** (1.34TB) and **OpenML benchmarks** (**PCA**, **leakage checks**, **type inference**); encoded invariances (hierarchical structure, numeric representation) that stabilized results across datasets.
- Implemented robust preprocessing and augmentation for mixed-type tables (numeric tokenization, text column embeddings, few-shot reference sampling); improved generalization to unseen schemas without metadata beyond column names.
- Built and maintained a rigorous experimentation pipeline using **Python** with **PyTorch** for baselines; reported reproducible R^2 and F1 metrics with best-checkpoint selection.
- Scaled training and evaluation on A100 GPUs with **HuggingFace Accelerate** and **Slurm**; reduced run-to-run variance and enabled zero-shot and few-shot experiments across seeds and datasets for reliable comparisons.

Undergraduate Research Scientist, *Vector Institute, Toronto, Ontario, Canada* May. 2024 - Aug. 2024
Supervisor: Pascal Poupart, University of Waterloo

- Developed a Contextual RBF-Histogram algorithm using **Python** for **contextual bandits** (RBF-kernel–weighted reward bins) to select the best arm per context, achieving 55.6% correct arm selection on a test set of 2K data points.
- Formulated and implemented a **contextual multi-armed bandit (CMAB)** procedure for sequential decision-making, using an RBF reward for greedy arm selection and **XGBoost** for label generation, with online data aggregation to improve the policy.
- Evaluated on datasets over 10K rows, producing diagnostic plots to measure uncertainty and explainability.

PUBLICATION

ERD: Extended RAW-Diffusion Framework for De-rendering sRGB Images Nov. 2025
Jiaqi Shang, *Yifan Qu*, Jianbo Qi
Abstract submitted to ISPRS 2026

Basis Transformers for Multi-Task Tabular Regression Jun. 2025
Wei Min Loh, **Jiaqi Shang**, Pascal Poupart
Submitted to AISTATS 2026 [Preprint]

PROJECT

ERD: Extended RAW-Diffusion Framework for De-rendering sRGB Images Sept. 2025 - Dec. 2025

- Developed an **extended RAW-Diffusion** model with **ControlNet**-guided conditioning for recovering sensor-space irradiance from sRGB images **across heterogeneous ISP pipelines**. Built an **end-to-end training and evaluation pipeline** on over 1000 images of the MIT-Adobe FiveK dataset with 7 camera models.
- Reduced adaptation cost by 60% and stabilized training by freezing a pre-trained RGB-guided diffusion backbone and an **efficient injection** of camera-conditioned residuals.

Determine Protein Information Flow Via Granger Causality Analysis Sept. 2025 - Dec. 2025

- Formulated **protein allostery mapping** as a causal inference problem on **molecular dynamics (MD) trajectories** of Adenylate Kinase (AdK) to infer **directed information flow** under **cyclic dependencies**.
- Developed a **neural Granger causality** framework to estimate residue-level information flow, combining a **Bayesian prior** to improve biological plausibility. Validated on **synthetic ground-truth simulations** and **real AdK trajectories**; recovered clean and directed signalling pathways consistent with known AdK mechanics and outperformed correlation-based baselines.

AWARD, SCHOLARSHIP, and GRANT

- ($7,500) Vector Institute Research Grant 2024
- ($1,500) University of Waterloo President's Research Award 2024
- ($3,000) University of Waterloo CS International Student Upper-Year Scholarship 2022
- ($2,000) University of Waterloo President's Scholarship of Distinction 2021

TECHNICAL SKILL

- Languages and Libraries: Python (PyTorch, NumPy, pandas, scikit-learn, SciPy; unittest/mock), R, C++, SQL, Bash.
- Platforms and DevOps Tools: Django, MySQL; Git and Bitbucket Pipelines (CI/CD); Docker, Kubernetes; AWS EC2.
- Machine Learning and Data: Transformers; NLP paraphrasing; safe/explainable RL (contextual bandits, Bayesian optimization); EDA and feature engineering; data augmentation.